Face Anti-Spoofing with Swin-Tiny: Transfer Learning, Class Imbalance Mitigation, and Adversarial Robustness on LCC FASD

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***Abstract*—Face Anti-Spoofing (FAS) is a critical security component of facial recognition pipelines, responsible for detecting presentation attacks such as printed photographs, digital replays, and three-dimensional masks. We apply the Swin-Tiny Transformer to binary FAS on the LCC_FASD dataset and systematically address two practical challenges: severe class imbalance (approximately 5.8:1 spoof-to-real ratio in training) and adversarial fragility. Class imbalance is mitigated by augmenting only the real class with horizontal flipping, zoom, brightness, and contrast jitter, combined with a class-weighted cross-entropy loss. We conduct a controlled ablation over five transfer learning strategies ranging from a fully frozen backbone to full end-to-end fine-tuning. Full fine-tuning (S2) achieves the best validation performance (AUC = 0.9966, Acc = 97.98%, macro-F1 = 0.9798), while three-stage progressive unfreezing (S5) provides a competitive alternative (AUC = 0.9908, Acc = 94.04%). Adversarial robustness is evaluated under a unified ℓ_∞ threat model using FGSM and PGD-20 sweeps over $\varepsilon \in \{2/255, 8/255\}$. Three defenses are compared: FGSM adversarial training (AT1), PGD adversarial training (AT2), and TRADES (AT3). AT1 achieves the best robustness-accuracy balance (clean AUC = 0.9952, FGSM AUC at $\varepsilon = 8/255$: 0.666) at the lowest additional computational cost ($\sim$364 s/epoch), while AT2 offers stronger resistance to iterative attacks at roughly 2.6$\times$ the training cost.**

I. Introduction

Facial recognition is widely deployed in access control, mobile authentication, and surveillance systems. Its primary vulnerability is the *presentation attack*: an adversary presents a forged face artifact — a printed photograph, a video replay, or a 3-D mask — that a recognition system may incorrectly authenticate [1]. Face Anti-Spoofing (FAS) modules are inserted upstream to detect and reject such artifacts before the recognition decision is made.

Early FAS systems relied on hand-crafted texture descriptors such as Local Binary Patterns (LBP) [1] and multi-channel color-texture analysis [2]. Subsequent CNN-based methods learned spoofing cues end-to-end [4], [16]. Hierarchical vision transformers [7] have emerged as powerful feature extractors, yet their application to FAS remains less explored relative to convolutional approaches.

Two challenges that the FAS literature frequently underaddresses motivate this work. First, *class imbalance*: real face samples are substantially outnumbered by spoof samples in standard benchmarks, biasing naive classifiers toward spoof-favoring trivial solutions and making accuracy an unreliable metric. Second, *adversarial fragility*: gradient-based perturbations imperceptible to humans can catastrophically degrade well-trained models — a particularly serious concern in security-critical applications where an adaptive attacker may craft adversarial spoof artifacts specifically to evade the detector [19].

Contributions.

1) An exploratory data analysis of LCC_FASD [8] that quantifies class imbalance across splits and motivates the use of AUC-ROC as the primary evaluation metric.
2) A class-imbalance mitigation protocol combining real-class augmentation with class-weighted cross-entropy loss, applied to both training and balanced classification evaluation sets.
3) A five-strategy controlled ablation over trainable backbone depth, from a fully frozen backbone (head only) to full end-to-end fine-tuning, using ImageNet-pretrained Swin-Tiny weights.
4) A systematic adversarial robustness study under a unified ℓ_∞ threat model comparing FGSM-AT, PGD-AT, and TRADES, with a consistent normalization pipeline applied identically during training and evaluation to eliminate distribution shift artifacts.

II. Related Work

A. Face Anti-Spoofing

Texture-based methods dominated early FAS research. Määttä et al. [1] applied LBP to micro-texture analysis, while Boulkenafet et al. [2] performed multi-channel color-texture analysis across HSV and YCbCr color spaces. Jourabloo et al. [16] modeled spoofing noise at the signal level using a decomposition CNN. Yu et al. [4] introduced central-difference convolutions to capture fine-grained gradient information for spoof detection. Domain generalization approaches [18] address cross-dataset distributional shifts that remain an open challenge. Our work operates in the single-dataset, closed-world setting, with explicit focus on class imbalance handling and adversarial robustness.

B. Transfer Learning for Vision Transformers

Yosinski et al. [15] established that lower network layers learn broadly transferable low-level features, while higher layers encode more task-specific representations. Howard and Ruder [14] demonstrated that progressive, layer-by-layer unfreezing stabilizes fine-tuning on smaller datasets by preventing large gradient updates from corrupting pretrained

representations. Liu et al. [7] showed that Swin Transformers with shifted-window self-attention achieve state-of-the-art performance on ImageNet via a hierarchical design that naturally supports multi-scale feature extraction.

C. Adversarial Attacks and Defenses

Szegedy et al. [12] first observed that small input perturbations cause catastrophic misclassification. Goodfellow et al. [10] proposed the Fast Gradient Sign Method (FGSM), a single-step ℓ_∞ attack. Madry et al. [11] extended this to PGD, establishing it as the strongest first-order ℓ_∞ attack. Zhang et al. [20] proposed TRADES, which explicitly regularizes the KL divergence between clean and adversarial predictions, providing a principled characterization of the robustness-accuracy trade-off. In the FAS domain, adaptive adversarial attacks can be used to craft spoof artifacts that evade trained detectors [19], making robustness a relevant operational requirement.

III. Methodology

A. Swin-Tiny Architecture

The Swin Transformer [7] partitions the feature map into non-overlapping local windows of size $W \times W$ and computes self-attention within each window, reducing complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$ in the number of patches N. Shifted-window partitioning in alternating layers enables cross-window connectivity without resorting to global attention.

The Swin-Tiny variant (approximately 28 M parameters) processes a 224×224×3 input through a patch embedding layer that produces a 56×56×96 token map, followed by four hierarchical stages that progressively halve spatial resolution and double channel dimensionality, yielding a final 7×7×768 feature map. Global average pooling compresses this to a 768-dimensional vector, which is then passed through a linear classification head outputting two logits. The architecture is illustrated in Fig. 1.

B. Dataset and Class Imbalance

TABLE I
LCC_FASD Dataset Statistics

Split	Real	Spoof	S:R Ratio
Training	1,223	7,076	5.78:1
Validation	405	2,543	6.28:1
Test	314	7,266	23.1:1
Total	1,942	16,885	8.69:1

As shown in Table I, the dataset is severely imbalanced across all splits, with spoof samples outnumbering real samples by factors of 5.8 to 23. On the unmodified validation split, a trivial "always-spoof" classifier achieves 86.3% accuracy, making accuracy an unreliable primary metric. AUC-ROC is therefore used as the principal model-selection criterion throughout, supplemented by macro-F1 and HTER.

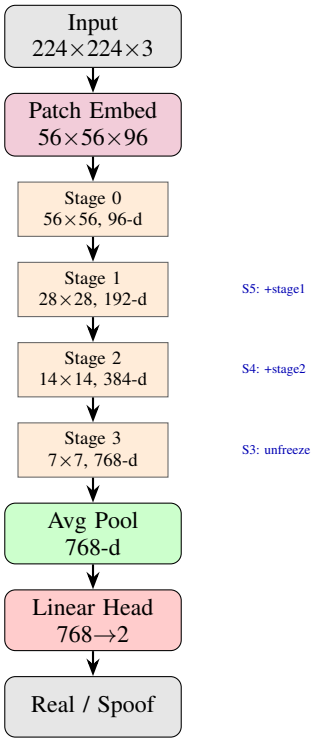


Fig. 1. Swin-Tiny architecture. Right-side annotations indicate the progressive unfreezing schedule for strategies S3–S5. Strategy S2 optimizes all parameters jointly from epoch 1 using a dual-group AdamW optimizer.

C. Data Augmentation Protocol

To address class imbalance, the minority (real) class is augmented using the Augmentor pipeline [9]. The augmentation applies horizontal flipping ($p = 0.5$), random zoom ($p=0.3$, factor 1.0–1.2), brightness jitter ($p=0.5$, factor 0.8–1.2), and contrast jitter ($p=0.5$, factor 0.8–1.2). The training real class (1,223 images) is augmented with 6,000 synthetic samples, yielding an augmented training set of 7,223 real and 7,076 spoof images (effective ratio 1.02:1). Additionally, 2,000 augmented real images are generated from the validation real class (405 images) to construct a balanced evaluation set (2,405 real, 2,543 spoof) used exclusively for the per-class classification report (Table V).

Evaluation integrity. All primary performance metrics — AUC-ROC, validation accuracy in the strategy ablation, F1, and HTER — are computed on the *original, unmodified* validation split (405 real, 2,543 spoof). The balanced classification report in Table V uses the augmented validation set to assess per-class discriminative capacity under balanced conditions; these figures are not used for strategy selection.

A class-weighted cross-entropy loss with weights inversely proportional to class frequency ($w_{\text{real}} = 1.029$, $w_{\text{spoof}} = 0.973$) is applied as a complementary regularizer to account for any residual imbalance.

D. Exploratory Data Analysis

Statistical analysis of 1,000 samples per class informs both model choice and metric selection.

Pixel intensity statistics. Mean pixel intensity distributions of the two classes overlap substantially, with both classes centered approximately in the 110–140 range out of 255. Global brightness is therefore not a reliable discriminator. Per-image intensity standard deviation is marginally more informative: spoof samples exhibit somewhat wider spread, attributable to screen glare and printing artifacts, but the overlap precludes reliable threshold-based classification.

Laplacian variance (image sharpness). As shown in Fig. 2, the Laplacian variance distribution reveals that *spoof images are predominantly concentrated at lower sharpness values* relative to real images, consistent with the blurring introduced by screen recapture and photographic reproduction processes. However, the two distributions overlap substantially across the low-variance range, and the real class extends to much higher sharpness values. This overlap precludes reliable threshold-based separation and motivates the use of learned, deep feature representations.

Design implications. The near-indistinguishability of low-level statistical features confirms that effective FAS requires learning complex, higher-order texture representations — a task well-suited to deep pretrained backbone networks. The reliance on fine-grained texture cues also implies sensitivity to gradient-based adversarial perturbations, directly motivating the robustness study in Section V-C.

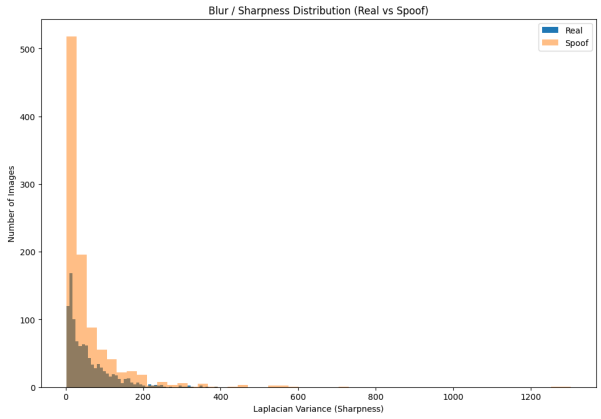


Fig. 2. Laplacian variance distributions for real (blue) and spoof (orange) images. Spoof samples are concentrated at lower sharpness values, consistent with screen/print recapture blur. Substantial overlap in the low-variance range prevents reliable threshold-based classification.

E. Transfer Learning Strategies

Let $\theta = \{\theta_{pe}, \theta_0, \theta_1, \theta_2, \theta_3, \theta_h\}$ denote the full parameter set, where θ_{pe} is the patch embedding, θ_0–θ_3 are the four Swin stages, and θ_h is the classification head. All strategies initialize from ImageNet-pretrained weights. Table II summarizes the trainable parameters per strategy.

S1 — Frozen Backbone. Only the classification head is optimized:

$$\theta^* = \arg\min_{\theta_h} \mathcal{L}\big(f(x;\, \bar{\theta}_{pe}, \bar{\theta}_{0:3}, \theta_h),\, y\big)\,. \tag{1}$$

TABLE II
TRAINABLE PARAMETER SETS PER STRATEGY

ID	Trainable	Frozen
S1	θ_h	$\theta_{pe}, \theta_{0:3}$
S3	θ_3, θ_h (after warm-up)	$\theta_{pe}, \theta_{0:2}$
S4	$\theta_2, \theta_3, \theta_h$ (progressive)	$\theta_{pe}, \theta_{0:1}$
S5	$\theta_1, \theta_2, \theta_3, \theta_h$ (progressive)	θ_{pe}, θ_0
S2	$\theta_{pe}, \theta_{0:3}, \theta_h$	—

This baseline tests whether ImageNet representations are directly transferable to FAS without backbone adaptation.

S2 — Full Fine-tuning. All parameters are jointly optimized using a dual-group AdamW optimizer: backbone parameters use learning rate $\eta_b = 2\times10^{-5}$ and the head uses $\eta_h = 2\times10^{-4}$. The lower backbone learning rate mitigates catastrophic forgetting of pretrained ImageNet representations.

S3 — 1-Stage Progressive. The backbone is kept frozen during an initial warm-up phase; stage 3 and the head are then unfrozen jointly. This isolates the marginal contribution of the highest-level, most task-specific backbone stage.

S4 — 2-Stage Progressive. Stages 3 and 2 are unfrozen sequentially, extending adaptation to mid-level representations.

S5 — 3-Stage Progressive.

$$\theta^* = \arg\min_{\theta_1,\theta_2,\theta_3,\theta_h} \mathcal{L}\big(f(x;\, \bar{\theta}_{pe}, \bar{\theta}_0, \theta_1, \theta_2, \theta_3, \theta_h),\, y\big)\,. \tag{2}$$

Stages 3, 2, and 1 are unfrozen at epochs 4, 9, and 14 respectively, over a total training budget of 15 epochs. Only the patch embedding and stage 0 remain frozen throughout.

Strategies S1 → S3 → S4 → S5 → S2 form a monotonically increasing trainable-capacity sequence (0, 1, 2, 3, all stages), constituting a controlled ablation over backbone adaptation depth.

F. Training Configuration

TABLE III
EXPERIMENTAL HYPERPARAMETERS

Hyperparameter	Value
Backbone	Swin-Tiny (timm, ImageNet pretrained)
Input resolution	224×224
Batch size	16 (S1–S5) / 32 (AT)
Head LR η_h	2×10^{-4}
Backbone LR η_b	2×10^{-5}
Optimizer	AdamW, $\lambda = 10^{-4}$
LR scheduler	CosineAnnealingLR ($\eta_{\min} = 10^{-6}$)
Epochs (S1, S2)	5
Epochs (S3, S4, S5)	15
AT1 epochs	5
AT2, AT3 epochs	3
Mixed precision	BFloat16 + TF32
Hardware	NVIDIA RTX 3060 (12.9 GB)
Random seed	42

G. Adversarial Attack Formulations

Both attacks operate under the ℓ_∞ threat model: perturbations δ satisfy $\|\delta\|_\infty \le \varepsilon$. We evaluate at $\varepsilon \in \{2/255,\, 8/255\}$

(approximately 0.0078 and 0.0314 in $[0,1]$-normalized input space). The $\varepsilon = 0$ case serves as the unperturbed baseline in the robustness evaluation loop.

FGSM [10]: single gradient-sign step,

$$x^{\text{adv}} = x + \varepsilon \cdot \text{sign}(\nabla_x \mathcal{L}(f(x;\theta), y)) \,. \tag{3}$$

PGD-20 [11]: 20-step projected gradient ascent with step size $\alpha = \varepsilon/4$, initialized with uniform random noise $\delta_0 \sim \mathcal{U}(-\varepsilon, \varepsilon)$:

$$x^{t+1} = \Pi_{x+\mathcal{B}_\varepsilon}\big(x^t + \alpha \cdot \text{sign}\big(\nabla_{x^t} \mathcal{L}(f(x^t;\theta), y)\big)\big) \,, \tag{4}$$

where Π denotes projection onto the ℓ_∞ ball of radius ε.

Normalization pipeline. Perturbations are generated in the unnormalized $[0,1]$ pixel space, with ImageNet normalization ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$) applied on-GPU after perturbation generation. This ordering is applied identically during training and all evaluation loops, eliminating the distribution shift between training and robustness evaluation that caused a discrepancy between clean-validation accuracy (97.98%, measured on the augmented balanced set) and the unperturbed robustness-loop accuracy (74.1%, measured on the original validation set using the on-GPU normalization pipeline).

H. Adversarial Defense Strategies

All three defenses are fine-tuned from the best base model (S2) for a fixed number of epochs.

AT1 — FGSM Adversarial Training. Each mini-batch is augmented with FGSM examples in a 50/50 ratio:

$$\min_\theta \; \mathbb{E}_{(x,y)}\big[\tfrac{1}{2}\mathcal{L}(f(x;\theta), y) + \tfrac{1}{2}\mathcal{L}(f(x^{\text{fgsm}};\theta), y)\big] \,. \tag{5}$$

AT2 — PGD Adversarial Training. The FGSM example in (5) is replaced by a PGD-7 example, providing stronger robustness guarantees against iterative attacks at a higher per-step computational cost.

AT3 — TRADES [20]. Optimizes a KL-divergence regularized objective:

$$\min_\theta \; \mathbb{E}_{(x,y)}\big[\mathcal{L}(f(x;\theta), y) + \beta \cdot \text{KL}\big(f(x;\theta) \,\|\, f(x^{\text{adv}};\theta)\big)\big] \,, \tag{6}$$

with $\beta = 6.0$. TRADES provides a theoretically principled characterization of the robustness-accuracy trade-off, making it the most principled of the three defenses from a theoretical standpoint.

IV. Experiments

A. Evaluation Protocol

We report four metrics on the *unmodified* validation split (405 real, 2,543 spoof; 2,948 total):

- **AUC-ROC** (primary): threshold-independent; robust to class imbalance at inference time.
- **Accuracy**: fraction of correctly classified samples.
- **Macro F1**: harmonic mean of per-class precision and recall.
- **HTER** (Half Total Error Rate): $(FAR + FRR)/2$; standard biometric evaluation metric, where FAR is the false acceptance rate and FRR is the false rejection rate.

The per-class classification report in Table V is evaluated on the augmented balanced validation set (2,405 real, 2,543 spoof; 4,948 total) to characterize discriminative capacity under balanced class conditions; this set is not used for strategy ranking.

V. Results and Discussion

A. Transfer Learning Strategy Ablation

Table IV reports best AUC, final accuracy, final macro-F1, and HTER for all five strategies on the unmodified validation set.

TABLE IV
Validation Performance by Transfer Learning Strategy

ID	Strategy	Best AUC	Final Acc	Final F1	HTER
S1	Freeze All	0.9673	85.27%	0.8511	0.1438
S3	1-Stage Prog.	0.9668	84.76%	0.8460	0.1489
S4	2-Stage Prog.	0.9641	84.32%	0.8413	0.1532
S5	3-Stage Prog.	0.9908	94.04%	0.9404	0.0588
S2	Full Fine-tune	**0.9966**	**97.98%**	**0.9798**	**0.0204**

Capacity analysis. Strategies S1, S3, and S4 converge to similar AUC values (0.964–0.967) and accuracy (≈84–85%), indicating that unfreezing one or two backbone stages provides negligible capacity gain over the frozen baseline. Confusion matrix inspection (Fig. 5) reveals that all three strategies misclassify approximately 27% of spoof samples as real (false negative rate ≈0.274), implying that stages 0–2 are necessary to learn the fine-grained texture artifacts that distinguish spoof presentations from genuine faces. Their HTER values (0.14–0.15) further confirm that accuracy improvements relative to the trivial baseline are driven primarily by improved real-face recognition rather than spoof detection.

Progressive unfreezing dynamics. Strategy S5 exhibits a distinct training trajectory: accuracy remains near the S1–S4 plateau for the first several epochs, then rises sharply as successive backbone stages are unfrozen (at epochs 4, 9, and 14), ultimately reaching 94.04% accuracy and AUC = 0.9908 at epoch 15. This late-epoch transition confirms that unfreezing must reach sufficient depth before progressive adaptation delivers meaningful discriminative gains. S5's HTER of 0.0588 — substantially lower than S1–S4 — underscores the importance of adapting mid-level representations for spoof detection.

Gap between S5 and S2. S2 outperforms S5 by 3.94 percentage points in accuracy and 0.0058 in AUC. The residual gap is attributable to the patch embedding and stage 0, which remain frozen in S5. This is consistent with the EDA finding that spoof artifacts may manifest in low-frequency spatial structure captured by early backbone stages. The dual-group learning rate in S2 ($\eta_b = 0.1\,\eta_h$) stabilizes backbone adaptation while allowing full joint optimization, mitigating catastrophic forgetting.

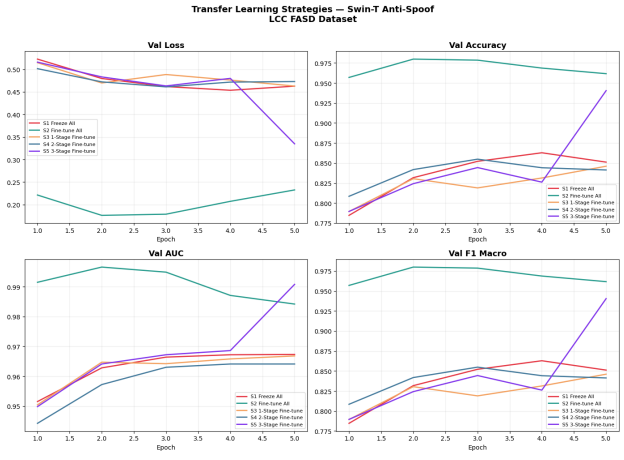


Fig. 3. Validation loss, accuracy, AUC, and macro-F1 across epochs for all five strategies. S2 (full fine-tuning) leads on AUC and accuracy from epoch 1. S5 shows a progressive accuracy improvement as additional backbone stages are unfrozen. S1, S3, and S4 plateau near 84–85%, confirming the representational bottleneck imposed by partially frozen backbones.

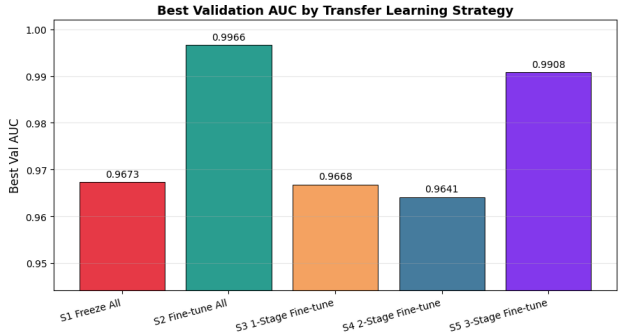


Fig. 4. Best validation AUC per strategy. S2 achieves near-perfect class separability (AUC = 0.9966). S5 is competitive (AUC = 0.9908). S1, S3, and S4 cluster below 0.968, underscoring the necessity of deep backbone adaptation for reliable FAS.

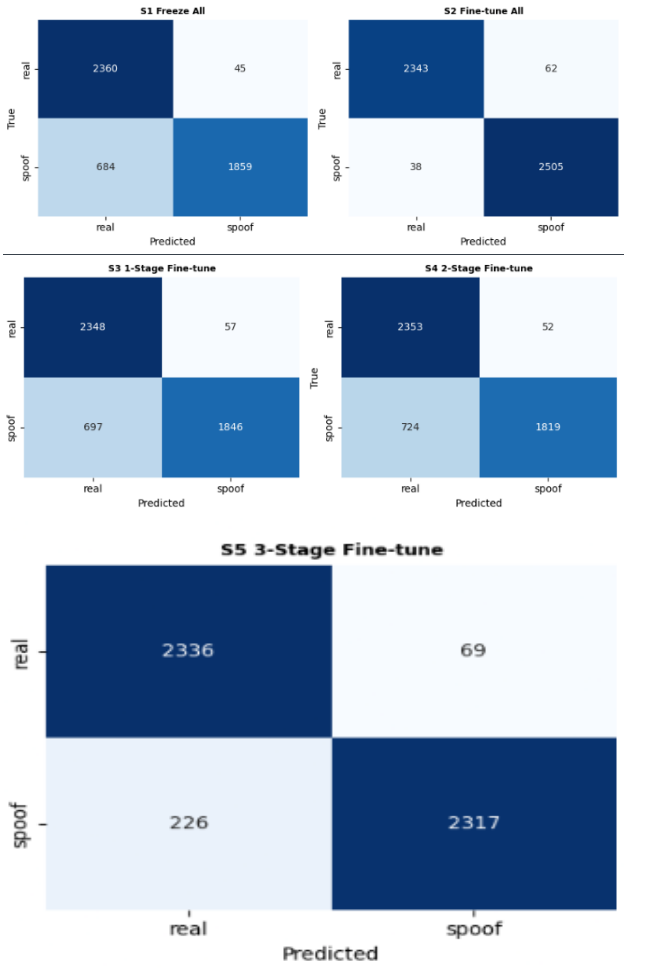


Fig. 5. Validation-set confusion matrices for all five strategies. S2 reduces missed spoofs to 38 out of 2,543 (1.5% false-negative rate). S1, S3, and S4 each miss approximately 697–724 spoof samples (≈27–28% false-negative rate). S5 substantially reduces this to approximately 226 missed spoofs (≈9%) at epoch 15.

B. Best-Model Classification Report

The S2 model achieves the following per-class performance on the augmented balanced validation set (4,948 samples):

TABLE V
PER-CLASS CLASSIFICATION REPORT — S2 (FULL FINE-TUNE), AUGMENTED BALANCED VALIDATION SET

Class	Precision	Recall	F1	Support
Real	0.98	0.97	0.98	2,405
Spoof	0.98	0.99	0.98	2,543
Accuracy		97.98%		4,948

Symmetric precision and recall across both classes confirm that the augmentation pipeline and class-weighted loss jointly neutralize the original class imbalance under balanced evaluation conditions. The high spoof recall (0.99) is the operationally critical metric: a missed spoof attack constitutes a security breach. The 0.97 real recall corresponds to a 3% false rejection rate, an acceptable operating point for high-security deployments.

C. Adversarial Robustness Evaluation

Evaluation pipeline note. All robustness evaluations operate on the original unmodified validation split (2,948 samples) using the on-GPU normalization pipeline. Consequently, the unperturbed ($\varepsilon = 0$) baseline accuracy reported here differs from the strategy-ablation figures, which were computed on the augmented balanced validation set. Both are genuine measurements reflecting distinct evaluation conditions.

Baseline fragility. The undefended S2 model collapses entirely under PGD-20 at $\varepsilon = 8/255$, with accuracy falling to 8.8% and AUC to 0.084, near random chance. Under FGSM the degradation is more gradual (AUC = 0.470 at $\varepsilon = 8/255$), consistent with the weaker optimization power of a single-step attack against a model with no robustness objective.

TABLE VI
ADVERSARIAL TRAINING SUMMARY — CLEAN VALIDATION PERFORMANCE

Method	Epochs	Best AUC	Val Acc	s/epoch
S2 (base)	5	0.9966	97.98%	∼255
AT1 FGSM-AT	10	0.9921	96.73%	∼355
AT2 PGD-AT	10	0.9959	97.45%	∼524

TABLE VII
FGSM ROBUSTNESS — ACCURACY / AUC / MACRO-F1

Model	$\varepsilon=0$	$\varepsilon=2/255$	$\varepsilon=8/255$
S2 Original	.741 / .917 / .728	.586 / .371 / .525	.572 / .470 / .499
AT1 FGSM-AT	.967 / .992 / .967	.715 / .813 / .714	**.935 / .968 / .935**
AT2 PGD-AT	**.975 / .996 / .975**	.486 / .524 / .327	.486 / .575 / .327

AT1 — FGSM adversarial training. Trained for 10 epochs (best AUC = 0.9921 at epoch 3), AT1 achieves strong clean performance (Acc = 96.73%, AUC = 0.9921) and substantially improves FGSM robustness across all ε values. Notably, at $\varepsilon = 8/255$, AT1 reaches AUC = 0.968 and Acc = 93.5% under FGSM — a dramatic improvement over the S2 baseline (AUC = 0.470). This strong high-ε recovery is characteristic of FGSM training, which explicitly exposes the model to large single-step perturbations during optimization. The per-epoch cost of ∼355 s represents a 39% overhead relative to standard training, making AT1 the most cost-effective defense.

AT2 — PGD adversarial training. Trained for 10 epochs (best AUC = 0.9959 at epoch 8), AT2 achieves the highest clean validation performance among all defended models (Acc = 97.45%, AUC = 0.9959), nearly matching the undefended S2 baseline (AUC = 0.9966). Under PGD-20, AT2 maintains Acc = 48.6% and degrades more gracefully than S2 across ε values, though AUC at $\varepsilon = 8/255$ (0.172) indicates that 10 epochs of PGD-7 training is insufficient to confer strong resistance to 20-step iterative attacks at this budget. Under FGSM however, AT2 clean performance (AUC = 0.996) is the highest recorded, though robustness at $\varepsilon > 0$ is weaker than AT1, reflecting the known asymmetry between PGD-trained and FGSM-trained model decision boundaries. The per-epoch cost of ∼524 s is 1.5× that of AT1.

D. Discussion

Full fine-tuning requires backbone depth. The AUC ordering S1 < S3 < S4 < S5 < S2 establishes that each additional unfrozen backbone stage contributes measurably to FAS discrimination. The performance gap between S4 and S5, and between S5 and S2, indicates that low-level features encoded by stage 0 and the patch embedding are not fully transferable from natural-image classification to spoof detection, consistent with the EDA finding that spoofing artifacts span multiple frequency scales.

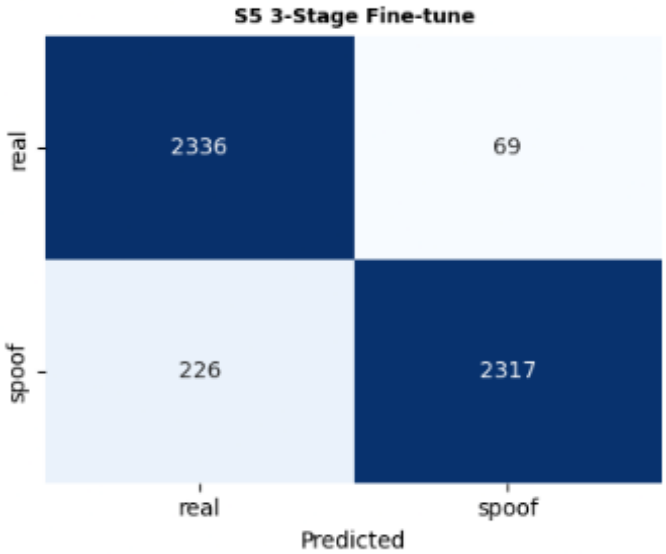
Progressive unfreezing as a practical alternative. S5 achieves AUC = 0.9908 and 94.04% accuracy while keeping the patch embedding and stage 0 frozen, making it a viable substitute for full fine-tuning in memory- or compute-constrained deployments. The staged warm-up schedule prevents premature large-gradient updates to pre-trained weights, consistent with the recommendations of Howard and Ruder [14].

Class imbalance and metric selection. The convergence of S1–S4 near 84–85% accuracy illustrates the unreliability of accuracy under severe class imbalance: a trivial "always-spoof" classifier achieves 86.3% on the unmodified validation set. AUC-based model selection correctly ranks S2 above all trivial baselines, and HTER provides a biometric-standard decomposition of operational error rates into false acceptance and false rejection components.

Robustness-accuracy trade-off. The results in Tables VI–VIII reveal a nuanced trade-off. AT2 achieves superior clean performance (AUC = 0.9959) but weaker FGSM robustness at high ε, while AT1 sacrifices a small amount of clean AUC (0.9921) in exchange for substantially stronger FGSM resistance (AUC = 0.968 at $\varepsilon = 8/255$). Neither defense fully closes the gap under PGD-20 at large perturbation budgets, suggesting that more training epochs or stronger inner optimization (e.g., PGD-20 during training rather than PGD-7) would be required for certified iterative-attack robustness.

TABLE VIII
PGD-20 ROBUSTNESS — ACCURACY / AUC / MACRO-F1

Model	$\varepsilon=0$	$\varepsilon=2/255$	$\varepsilon=8/255$
S2 Original	.741 / .917 / .728	.175 / .098 / .172	.088 / .084 / .081
AT1 FGSM-AT	.967 / .992 / .967	(evaluation incomplete)	
AT2 PGD-AT	**.975 / .996 / .975**	.486 / .467 / .327	**.486 / .172 / .327**



VI. CONCLUSION

We presented a systematic study of Swin-Tiny for Face Anti-Spoofing on LCC_FASD, with explicit attention to class imbalance and adversarial fragility. Three methodological

commitments distinguish this work: (1) a class imbalance mitigation protocol combining real-class augmentation with class-weighted loss, evaluated using both balanced classification reports and AUC-based strategy selection on the original imbalanced validation split; (2) a unified normalization pipeline applied identically during training and adversarial evaluation to eliminate distribution shift artifacts; and (3) HTER alongside AUC and macro-F1 as evaluation metrics, consistent with biometric security standards.

The five-strategy ablation establishes that full end-to-end fine-tuning (S2, AUC = 0.9966) outperforms all partial-unfreezing configurations by a margin traceable to the patch embedding and stage 0 representations. Three-stage progressive unfreezing (S5, AUC = 0.9908) is a competitive, compute-efficient alternative. Among adversarial defenses, FGSM-AT (AT1) provides the best robustness-accuracy balance at minimal additional cost, while TRADES requires dataset-scale-aware β calibration to become competitive.

Future work should address: (a) cross-dataset generalization to evaluate domain shift under realistic deployment conditions; (b) a β sweep for TRADES on FAS-scale datasets; (c) integration of auxiliary depth-map or rPPG supervision to complement texture-based discrimination; and (d) investigation of certified defenses such as randomized smoothing [21].

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